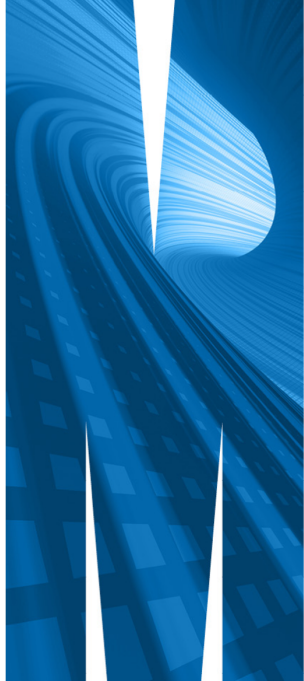


ETC3550/ETC5550

Applied forecasting

Week 2: Time series graphics



Outline

- 1 dplyr functions
- 2 Time series graphics
- 3 Seasonality and cyclicity
- 4 White noise and random walks
- 5 Lag plots and ACFs

Outline

- 1 dplyr functions
- 2 Time series graphics
- 3 Seasonality and cyclicity
- 4 White noise and random walks
- 5 Lag plots and ACFs

dplyr functions

- `filter()`: choose rows
- `select()`: choose columns
- `mutate()`: make new columns
- `group_by()`: group rows
- `summarise()`: summarise across groups

Outline

- 1 dplyr functions
- 2 Time series graphics
- 3 Seasonality and cyclicity
- 4 White noise and random walks
- 5 Lag plots and ACFs

Time series graphics

- Time plots: `autoplot()`
- Seasonal plots: `gg_season()`
- Seasonal subseries plots: `gg_subseries()`
- Lag plots: `gg_lag()`
- ACF plots: `ACF() |> autoplot()`

Outline

- 1 dplyr functions
- 2 Time series graphics
- 3 Seasonality and cyclicity
- 4 White noise and random walks
- 5 Lag plots and ACFs

Time series patterns

Trend pattern exists when there is a long-term increase or decrease in the data.

Seasonal pattern exists when a series is influenced by seasonal factors (e.g., the quarter of the year, the month, or day of the week).

Cyclic pattern exists when data exhibit rises and falls that are *not of fixed period* (duration usually of at least 2 years).

Time series components

Differences between seasonal and cyclic patterns:

- seasonal pattern constant length; cyclic pattern variable length
- average length of cycle longer than length of seasonal pattern
- magnitude of cycle more variable than magnitude of seasonal pattern

Time series components

Differences between seasonal and cyclic patterns:

- seasonal pattern constant length; cyclic pattern variable length
- average length of cycle longer than length of seasonal pattern
- magnitude of cycle more variable than magnitude of seasonal pattern

The timing of peaks and troughs is predictable with seasonal data, but unpredictable in the long term with cyclic data.

Outline

- 1 dplyr functions
- 2 Time series graphics
- 3 Seasonality and cyclicity
- 4 White noise and random walks**
- 5 Lag plots and ACFs

White noise and random walks

White noise

$\varepsilon_t \sim$ independent and identically distributed with mean zero and constant variance.

Random walks

$$y_t = y_{t-1} + \varepsilon_t$$

where ε_t is a white noise variable.

Outline

- 1 dplyr functions
- 2 Time series graphics
- 3 Seasonality and cyclicity
- 4 White noise and random walks
- 5 Lag plots and ACFs

Sampling distribution of autocorrelations

Sampling distribution of r_k for white noise data is asymptotically $N(0, 1/T)$.

Sampling distribution of autocorrelations

Sampling distribution of r_k for white noise data is asymptotically $N(0, 1/T)$.

- 95% of all r_k for white noise must lie within $\pm 1.96/\sqrt{T}$.
- If this is not the case, the series is probably not WN.
- Common to plot lines at $\pm 1.96/\sqrt{T}$ when plotting ACF. These are the **critical values**.

Trend and seasonality in ACF plots

- When data have a trend, the autocorrelations for small lags tend to be large and positive.
- When data are seasonal, the autocorrelations will be larger at the seasonal lags (i.e., at multiples of the seasonal frequency)
- When data are trended and seasonal, you see a combination of these effects.